

LAURA BRAVO-SÁNCHEZ

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RESEARCH INTERESTS

Computer vision for analyzing human social interaction and behavior, focusing on 3D reconstruction and video understanding in real-world scenes.

EDUCATION

PhD., Biomedical Data Science, Stanford University	2021 - 2026
MSc., Biomedical Engineering, Universidad de los Andes	2017 - 2019
BSc., Biomedical Engineering, Universidad de los Andes	2013 - 2017
Minor in French Language and Culture	

WORK EXPERIENCE

Naver Labs Europe - Research Intern - Supervisor: Fabien Baradel	04/2025 - 09/2025
Universidad de los Andes - Researcher - Supervisor: Pablo Arbeláez	2019 - 2021
T&G S.A.S - Research Scientist - Supervisor: Carlos E. Pérez	02/2019 - 05/2019

SELECTED PUBLICATIONS

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- **All-age Human Mesh Recovery**
L. Bravo-Sánchez, et al.
CVPR Findings (2026)
 - **Human Mesh Modeling for Anny Body**
R. Brégier, G. Fiche, L. Bravo-Sánchez, et al.*
Arxiv (2025)
 - **MicroVQA: A Multimodal Reasoning Benchmark for Microscopy-Based Scientific Research**
L. Bravo-Sánchez, J. Burgess*, J.J. Nirschl*, et al.*
Proceedings of the Computer Vision and Pattern Recognition Conference (2025)
 - **Automating Maternal Entropy Calculation in Parent-Child Interactions Using Computer Vision**
Z. Wang, L. Bravo-Sánchez, et al.
Under review at Scientific Reports
 - **Ask, pose, unite: Scaling data acquisition for close interactions with vision language models.**
L. Bravo-Sánchez, J. Heo, Z. Weng, K.C. Wang, S. Yeung-Levy
Emergent Visual Abilities and Limits of Foundation Models CVPRW (2025)
 - **Artificial intelligence-powered 3D analysis of video-based caregiver-child interactions**
Z. Weng, L. Bravo-Sánchez, et al.
Science Advances (2025)
 - **Diffusion-HPC: Synthetic data generation for human mesh recovery in challenging domains.**
Z. Weng, L. Bravo-Sánchez, & S. Yeung-Levy
International Conference on 3D Vision (2024)
 - **Smart pooling: AI-powered COVID-19 informative group testing.**
M. Escobar, G. Jeanneret, L. Bravo-Sánchez, et al.
Scientific Reports (2022)

- **Surgical instrument grounding for robot-assisted interventions.**
*L. Bravo-Sánchez**, *C. González** & *P. Arbeláez*
Computer Methods in Biomechanics and Biomedical Engineering: Imaging (2022)
- **ISINet: An Instance-Based Approach for Surgical Instrument Segmentation.**
*L. Bravo-Sánchez**, *C. González** & *P. Arbeláez*
Medical Image Computing and Computer Assisted Intervention (2020)
- **Finding Four-Leaf Clovers: A Benchmark for Fine-Grained Object Localization.**
*L. Bravo-Sánchez**, *A. Pardo**, *G. Perez**, *P. Arbeláez*
Sixth Workshop on Fine-Grained Visual Categorization, CVPR (2019).

* denotes equal contribution.

AWARDS AND TALKS

- “How can Computer Vision guide the understanding of parent-child interactions?”. Technical talk at the 2024 WiDS Worldwide conference.
- Fulbright Colombia Minciencias Scholarship recipient 2021 Cohort (3 % acceptance, \$80.000 USD).
- Leader of Team Uniandes in the MISAW challenge part of MICCAI 2020. Won first place in the Activity Recognition task.
- Leader of Team Uniandes. Winers of 5 awards at the Robust Endoscopic Instrument Segmentation Challenge 2019 part of MICCAI 2019.
- “Totæ Lacrimæ: automatic recognition of human emotions based on micrographs of tear crystals”. Art exhibition (2019).

TEACHING EXPERIENCE

Graduate Teaching Assistant - Stanford University	<i>Spring 2023, 2024</i>
Graduate Teaching Assistant - Universidad de los Andes	<i>Fall 2017, 2018</i>
Undergraduate Teaching Assistant - Universidad de los Andes	<i>Fall 2016 - Spring 2017</i>

SERVICE AND OUTREACH

• Volunteer interpreter - Immigrants' Rights Clinic, Stanford University	<i>2024 - 2025</i>
• Student representative - DBDS, Stanford University	<i>2023 - 2024</i>
• Financial officer - Colombian Association, Stanford University	<i>2023 - 2024</i>
• Mentor - Innovation Girls 4.0 program - Visible Hands Corporation	<i>2015, 2020</i>
• Volunteer teacher - ColombiaCrece	<i>2018</i>
• Volunteer - Techo Colombia	<i>2012 - 2014</i>